VAISHNAVI PATIL

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EDUCATION

Northeastern University, Masters in ECE (Microsystem, Materials and Devices), Boston MA (CGPA: 3.9) **May 2026**

Coursework: Analog Integrated Circuits & Design, MEMS Design, VLSI Design, Solid States and Devices, Thin Film Technologies, Micro-nano Fabrication, Wireless Sensor/IOT, Fundamentals of Computer Engineering

Dayananda Sagar College of Engineering, BE in Electronics and Communication, India (8.0/10.0) **Jul 2024**

Coursework: VLSI, System Verilog, ASIC Design, Analog Electronics, Digital Electronics, FPGA, Microcontrollers

TECHNICAL SKILLS

Hardware & Circuit Design: Schematic design, SPICE Simulations, Op-amps, DC/DC Converters, data converters, analog signal
Digital Design: Verilog, SystemVerilog, RTL Design, FSM Development, AHB/APB Protocols, Timing closure, SoC Integration
Tools and Equipment: Cadence Virtuoso, Xilinx Vivado, ModelSim, Oscilloscope, Multimeter, Soldering, PSPICE, KiCAD
Programming Languages: C, C++, Python, Verilog, VHDL, Perl, LabVIEW (Familiar), Silvaco TCAD
Semiconductor Manufacturing & Process Engineering: IC Packaging, Wafer Processing, Wafer Probe & Final Testing

WORK EXPERIENCE

Backend VLSI Engineer, **QBit Semiconductor LTD** **May 2025 – Aug 2025**

- Executed full **RTL-to-GDSII** implementation of a **32-bit RISC-V** processor in **28nm CMOS** using **Cadence Genus,** Innovus, and PrimeTime, achieving 1.0 GHz with clean signoff across synthesis, P&R, and timing closure.
- Synthesized the **RISC-V RTL** in **Cadence Genus** with timing, power and area-driven optimization, achieving **1.0 GHz** and generating an area-efficient netlist for physical design handoff.
- Developed floorplan and power grid architecture with optimized macro placement and congestion-aware routing, achieving 68% core utilization and improved routability within constrained die area.
- Performed placement optimization, legalization, and congestion analysis, reducing routing overflow by 22% through hotspot mitigation, buffer insertion, and density-aware placement techniques.
- Implemented **CTS** and routing to meet 1.0 GHz timing, achieving 80ps clock skew while resolving all DRC, antenna, and connectivity violations with clean **Calibre DRC/LVS** signoff.
- Performed multi-corner **STA** in **PrimeTime across SS/TT/FF corners**, reducing timing violations by 30% through **ECO fixes** and **Tcl automation**, achieving positive **WNS/TNS** closure and 25% faster backend runtime.

*Digital Design Engineer***, Micron Technology** **Oct 2023 – Jul 2024**

- Designed a **16×16 systolic-array CNN accelerator** in **Verilog/SystemVerilog** with pipelined **MAC PEs,** achieving 2.4 TOPS at 900 MHz in 28nm CMOS for area- and power-constrained edge **AI workloads.**
- Developed parameterized **RTL** modules for control FSMs, buffering, and memory scheduling, reducing external memory access overhead by 32% and improving **CNN** accelerator efficiency.
- Designed self-checking SystemVerilog testbenches in **Cadence Xcelium,** validating **MAC** array and control logic across edge cases and resolving **15+ RTL** mismatches before synthesis handoff.
- Synthesized the accelerator in **Cadence Genus** with timing-driven **SDC** constraints, achieving 900 MHz timing closure with +0.11 ns WNS and 18% lower critical path delay through logic and cell optimization.
- Defined multicycle paths, clock gating, and I/O delay constraints in SDC, eliminating false timing violations and enabling clean PD handoff with zero constraint mismatches across SS/TT/FF corners.
- Automated the **RTL-to-synthesis flow** using **Python** scripting to eliminate manual steps across multiple PVT corners, reducing design iteration runtime by 25% and enabling consistent, repeatable timing closure for the full accelerator design.

Physical Design Engineer Intern, **Centum Electronics** **Apr 2023 - Sep 2023**

- Assisted in the development of the complete **RTL-to-GDSII** physical design flow for high-speed digital blocks in automotive grade SoCs. Ensured all processes complied with ISO 26262 standards for reliability and performance.
- Optimized floorplanning, power planning, and placement for **DDR4 memory subsystems** to achieve **3200 MT/s** throughput at 1.2 V. Balanced area, timing, and power trade-offs to meet design specifications effectively.
- Supported Clock Tree Synthesis (CTS) and routing processes by identifying and resolving skew and insertion delay challenges. Ensured robust clock domain integrity throughout the design.
- Conducted Static Timing Analysis (STA) to perform timing closure by identifying and fixing setup and hold violations. Ensured timing robustness across multiple Process, Voltage and Temperature **(PVT) corners.**
- Collaborated with **DRC/LVS** verification teams to identify and fix design rule and connectivity violations. Ensured the design met signoff criteria and was ready for tapeout.
- Collaborated with RTL, synthesis, and physical verification teams to improve backend design efficiency.

Design Verification Intern, **Maven Silicon** **Oct 2022 - Dec 2022**

- Designed and verified an **AHB2APB bridge** using Verilog, facilitating seamless communication between high-performance AHB masters and low-power APB peripherals in SoC architectures.
- Developed a 3-state FSM to control AHB transfer phases (IDLE, SETUP, ENABLE) and ensured precise mapping of control signals for APB protocol compliance.
- Designed key components such as **address decoder, control logic, write/read data path**, and **ready/valid handshake mechanisms** to enable correct bridging between the two protocols.
- Built a synthesizable top-level module by instantiating interface wrappers and managing signal synchronization between AHB and APB domains. Ensured seamless communication across bus protocols.
- Created a testbench environment using **ModelSim** for functional simulation and waveform analysis; validated transfer integrity, latency, and protocol correctness across corner cases.
- Debugged timing mismatches and edge-triggered failures during simulation using signal-level waveform inspection and refined the bridge **FSM** to ensure reliable data transactions.

PROJECTS

Design and Physical Implementation of CMOS XOR Gate in 180nm Technology **Jan 2026 - May 2026**

- Designed and implemented a **CMOS XOR** gate at transistor level using **Cadence Virtuoso** in **180nm** CMOS technology, including schematic capture, custom layout design, **DRC/LVS** verification, and parasitic extraction for accurate functional validation.
- Generated the final **GDSII** file for fabrication by optimizing transistor placement, routing, and layer connectivity while ensuring compliance with semiconductor fabrication design rules and improving layout efficiency.
- Performed post-layout simulations and fabrication-aware analysis to evaluate timing, power, and signal integrity, enabling successful physical verification and tape-out readiness for **CMOS** fabrication flow.

32 Bit ALU in 180nm CMOS Using Cadence Virtuoso. **Sep 2025 - Dec 2025**

- Designed a 32-bit ALU at the transistor level in 180nm CMOS using Cadence Virtuoso by developing and simulating custom schematics for functional blocks (adder, multiplexer, logic unit, comparator) to support arithmetic and logical operations.
- Integrated modules into a complete datapath and executed layout design with proper well tap insertion, DRC/LVS compliance, and metal routing optimization.
- Executed pre- and post-layout simulations with Spectre and optimized the design to meet area, power, and timing requirements within 180nm CMOS technology constraints.

Evolution of Gate Oxide Technology in MOSFET Devices **Aug 2025 – Dec 2025**

- Conducted a comprehensive study on the evolution of gate oxide technology in **MOSFET** devices, analyzing the transition from conventional **SiO_2** dielectrics to advanced high-k materials for nanoscale **CMOS** applications.
- Investigated gate oxide scaling challenges including leakage current, short-channel effects, oxide reliability, and tunneling phenomena, evaluating their impact on transistor performance, power consumption, and device reliability.
- Analyzed modern gate dielectric engineering techniques such as **high-k/metal gate integration** and **future oxide technologies** to understand their role in improving switching performance, scalability, and semiconductor device efficiency.

Physical Design of 741 Operational Amplifier in 180nm CMOS. **Jan 2025 - Apr 2025**

- Designed and implemented the full custom layout of a 741 Op-Amp in 180nm CMOS using Cadence Virtuoso, ensuring DRC/LVS compliance and verifying performance through post-layout simulations.
- Conducted **pre- and post-layout simulations** using Spectre to analyze open-loop gain, slew rate, and bandwidth, verifying layout impact on performance.
- Optimized the layout to minimize area and parasitic capacitance while maintaining signal integrity, ensuring stable and robust analog performance across PVT variations.

Design and Analysis of FDSOI CMOS **Sep 2024 - Dec 2024**

- Designed and simulated digital CMOS circuits using **22nm** Fully Depleted Silicon-On-Insulator (FDSOI) technology, leveraging body-biasing techniques for dynamic threshold voltage control and improved low-power operation.
- Analyzed key performance metrics including propagation delay, leakage current, dynamic power consumption, and noise immunity, comparing FDSOI-based designs with conventional bulk CMOS architectures for enhanced energy efficiency.
- Conducted **SPICE-level** simulations and process variation analysis to evaluate scalability, reliability, and performance optimization, highlighting the advantages of **FDSOI** technology for advanced sub-28nm semiconductor applications.z